

LAB 4:

Aim : To implement and execute advanced SQL queries on a relational database using joins, subqueries, set operations, aggregate functions, grouping, and conditional operators for effective data retrieval and analysis.

Q1. Find match Ids of all the matches in the database in which Tendulkar batted.

```
mysql> SELECT MatchID FROM Batting, Player;
```

```
mysql> SELECT MatchID FROM Batting,Player WHERE PID=PlayerID;
```

```
mysql> SELECT MatchID FROM Batting,Player WHERE PID=PlayerID AND  
Lname='Tendulkar';
```

OR

```
mysql> SELECT MatchID FROM Batting WHERE PID IN (SELECT  
PlayerID FROM Player WHERE LName='Tendulkar'); +-----+
```

```
MySQL localhost:3306 ssl anannya_match SQL > SELECT MatchID FROM Batting,Player WHERE PID=PlayerID;
```

MatchID
2755
2755
2755
2755
2755
2689
2689
2689
2689
2689
2689
2689
2755
2689
2755

```
16 rows in set (0.0006 sec)
```

```
MySQL localhost:3306 ssl anannya_match SQL > SELECT MatchID FROM Batting,Player WHERE PID=PlayerID AND Lname='Tendulkar';
```

MatchID
2689

Q2. Find match information of all matches in which Dohni has batted.

```
mysql> SELECT Team1,Team2,Ground,Date FROM ODIMatch,Batting,Player;
```

```
mysql> SELECT Team1,Team2,Ground,Date FROM ODIMatch WHERE MatchID IN  
(SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM Player  
WHERE LName ='Dhoni'));
```

OR

Using Join

```
mysql> SELECT Team1,Team2,Ground,Date FROM ODIMatch,Batting,Player WHERE
ODIMatch.MatchID=Batting.MatchID AND PlayerID=PID AND LName='Dhoni';
```

```
MySQL localhost:3306 ssl anannya_match SQL > SELECT Team1,Team2,Ground,Date FROM ODIMatch WHERE MatchID IN (SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM Player WHERE LName ='Dhoni'));
```

Team1	Team2	Ground	Date
Sri Lanka	India	Colombo	2008-08-24
Australia	India	Brisbane	2008-03-04

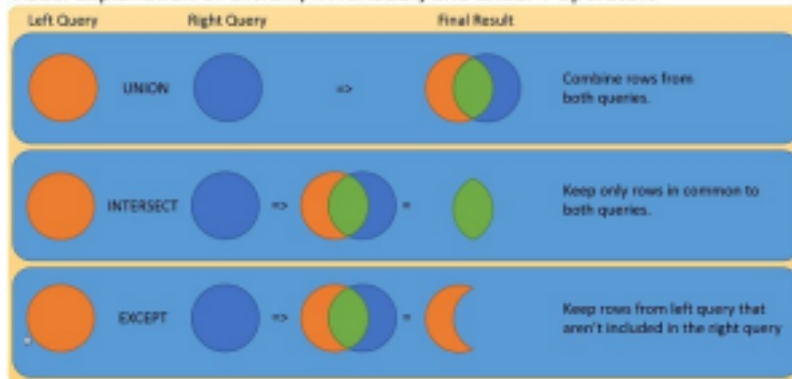
Q3. Find match Ids of all matches in which Tendulkar batted.

```
mysql> SELECT MatchID FROM Batting WHERE PID IN (SELECT
PlayerID FROM Player WHERE Lname='Tendulkar'); OR
```

Using Join

```
mysql> SELECT MatchID FROM Batting,Player WHERE
PID=PlayerID AND LName='Tendulkar';
```

Visual Explanation of UNION, INTERSECT, and EXCEPT operators



```
MySQL localhost:3306 ssl anannya_match SQL > SELECT MatchID FROM Batting,Player WHERE PID=PlayerID AND LName='Tendulkar';
```

MatchID
2689

Q4. Find Ids of all players that have both bowled and batted in the ODI match 2689

```
mysql> SELECT PID FROM Batting WHERE MatchID=2689 AND PID
```

IN (SELECT PID FROM Bowling WHERE MatchID=2689);

```
MySQL localhost:3306 ssl anannya_match SQL > SELECT PID FROM Batting
WHERE MatchID=2689 AND PID IN (SELECT PID FROM Bowling WHERE MatchID=26
89);
```

PID
99001
24001
23001
98002

Q5. Find Ids of all players that have either bowled or batted (or did both) in ODI match 2689

mysql> SELECT PID FROM Batting WHERE MatchID=2689 OR PID IN
(SELECT PID FROM Bowling WHERE MatchID=2689);

Using Union

mysql> SELECT PID FROM Batting WHERE MatchID=2689 UNION
(SELECT PID FROM Bowling WHERE MatchID=2689); +-----+

```
MySQL localhost:3306 ssl anannya_match SQL > SELECT PID FROM Batting
WHERE MatchID=2689 UNION (SELECT PID FROM Bowling WHERE MatchID=2689);
```

PID
89001
23001
25001
99002
95001
24001
99001
27001
98002

Q6. Find Ids

mysql> SELECT PID FROM Batting WHERE MatchID=2689 AND PID NOT
IN (SELECT PID FROM Bowling WHERE MatchID=2689); +-----+

```
MySQL localhost:3306 ssl anannya_match SQL > SELECT PID FROM Batting
WHERE MatchID=2689 AND PID NOT IN (SELECT PID FROM Bowling WHERE MatchI
D=2689);
```

PID
89001
25001
99002
95001
27001

Q7. Find match Ids where Tendulkar has played.

```
mysql> SELECT MatchID FROM Batting,(SELECT PlayerID FROM Player
WHERE LName='Tendulkar') ST WHERE PID=ST.PlayerID; +-----+
```

```
MySQL localhost:3306 ssl anannya_match SQL > SELECT MatchID FROM Bat
ting,(SELECT PlayerID FROM Player WHERE LName='Tendulkar') ST WHERE PID=
ST.PlayerID;
+-----+
| MatchID |
+-----+
|    2689 |
+-----+
```

Q8. Find match Ids of all those matches in which player 27001 bats and makes more runs than he made at every match he played at Colombo

```
mysql> SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns > ALL (SELECT
NRuns FROM Batting WHERE PID=27001 AND MatchID IN (SELECT MatchID FROM
ODIMatch WHERE Ground='Colombo'));
```

```
MySQL localhost:3306 ssl anannya_match SQL > SELECT MatchID FROM Ba
tting WHERE PID=27001 AND NRuns > ALL (SELECT NRuns FROM Batting WHERE P
ID=27001 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Col
ombo'));
```

```
+-----+
| MatchID |
+-----+
|    2689 |
+-----+
```

Q8. Find match Ids of all those matches in which player 27001 has scored minimum runs than he made at every match he played at Brisbane

```
mysql> SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns < ALL (SELECT
NRuns FROM Batting WHERE PID=27001 AND MatchID IN (SELECT MatchID FROM
ODIMatch WHERE Ground='Brisbane'));
```

```
MySQL localhost:3306 ssl anannya_match SQL > SELECT MatchID FROM Ba
tting WHERE PID=27001 AND NRuns < ALL (SELECT NRuns FROM Batting WHERE P
ID=27001 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Bri
sbane'));
```

```
+-----+
| MatchID |
+-----+
|    2755 |
+-----+
```

Q9(a). Find Ids and scores of players who scored less than 75 but more than 50 in Colombo.

```
mysql> SELECT PID,NRuns FROM Batting WHERE NRuns BETWEEN 51 AND
74 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE
Ground='Colombo');
```

```
MySQL localhost:3306 ssl anannya_match SQL > SELECT PID,NRuns FROM
Batting WHERE NRuns BETWEEN 51 AND 74 AND MatchID IN (SELECT MatchID FRO
M ODIMatch WHERE Ground='Colombo');
```

PID	NRuns
25001	71
91001	60

Q9(b). Find Ids and scores of players who scored greater than 49 runs in Colombo.

```
mysql> SELECT PID,NRuns FROM Batting WHERE NRuns > 49 AND MatchID IN
(SELECT MatchID FROM ODIMatch WHERE Ground='Colombo');
```

```
MySQL localhost:3306 ssl anannya_match SQL > SELECT PID,NRuns FROM
Batting WHERE NRuns > 49 AND MatchID IN (SELECT MatchID FROM ODIMatch WH
ERE Ground='Colombo');
```

PID	NRuns
25001	71
91001	60

Q10. Find the player Ids of those players whose date of first match (FTest) is not given in the database.

```
MySQL localhost:3306 ssl anannya_match SQL > SELECT PlayerID FROM P
layer WHERE FTest IS NULL;
```

PlayerID
27001

```
mysql> SELECT PlayerID FROM Player WHERE FTest IS NULL;
```

Q11. Find names of the player who batted in match 2689

```
mysql> SELECT FName,LName FROM Player WHERE EXISTS ( SELECT *
```

```
FROM Batting WHERE PlayerID=PID AND MatchID=2689); OR
```

```
mysql> SELECT FName,LName FROM Player,Batting WHERE PlayerID = PID AND
```

MatchID=2689;

```
MySQL localhost:3306 ssl anannya_match SQL > SELECT FName,LName FROM
M Player,Batting WHERE PlayerID = PID AND MatchID=2689;
```

FName	LName
Sachin	Tendulkar
Yuvraj	Singh
MS	Dhoni
Adam	Gilchrist
Ricky	Ponting
Andrew	Symonds
Breet	Lee
Praveen	Kumar
Harbhajan	Singh

Q12. Find player Ids of players who have scored more than 30 in every ODI match that they have played.

mysql> SELECT PID AS PlayerID FROM Batting b1 WHERE NOT EXISTS (SELECT *
FROM Batting b2 WHERE b1.PID=b2.PID AND NRuns < 31); +-----+

```
MySQL localhost:3306 ssl anannya_match SQL > SELECT PID AS PlayerID
FROM Batting b1 WHERE NOT EXISTS (SELECT * FROM Batting b2 WHERE b1.PID
=b2.PID AND NRuns < 31);
```

PlayerID
24001
25001
25001
89001
91001

Q13. Find the names of players that have batted in all the ODI matches in Melbourne that are in the database.

```
SELECT FName, LName
FROM Player
WHERE NOT EXISTS
(SELECT *
FROM ODIMatch
WHERE Ground = 'Melbourne'
AND NOT EXISTS
(SELECT *
FROM Batting
WHERE PID=PlayerID
AND (Batting.MatchId=ODIMatch.MatchID));
```

29 rows in set (0.0007 sec)

MySQL localhost:3306 ssl anannya_match SQL > select * from ODIMatch where Ground = "Melbourne";

MatchID	Team1	Team2	Ground	Date	Winner
2675	Australia	India	Melbourne	2008-02-10	Team2

Q14. Find the number of players that bowled in ODI match 2689

mysql> SELECT COUNT(*) AS NBowlers FROM Bowling WHERE MatchID=2689;

MySQL localhost:3306 ssl anannya_match SQL > SELECT COUNT(*) AS NBowlers FROM Bowling WHERE MatchID=2689;

NBowers
4

Q15. Find the average batting score of all the players that batted in the ODI Match 2689

mysql> SELECT AVG(NRuns) AS AvgRuns_2689 FROM Batting WHERE MatchID=2689;

MySQL localhost:3306 ssl anannya_match SQL > SELECT AVG(NRuns) AS AvgRuns_2689 FROM Batting WHERE MatchID=2689;

AvgRuns_2689
25.2222

Q16. Find the youngest player in the database

mysql> SELECT FName AS Youngest_Player FROM Player
WHERE YBorn=(SELECT MAX(YBorn) FROM Player);

MySQL localhost:3306 ssl anannya_match SQL > SELECT FName AS Youngest_Player FROM Player WHERE YBorn=(SELECT MAX(YBorn) FROM Player);

Youngest_Player
Ishant

Q17. Find the number of players in the database from each country

mysql> SELECT Country, COUNT(*) AS NPlayers FROM Player GROUP BY Country;

```
MySQL localhost:3306 ssl anannya_match SQL > SELECT Country, COUNT(*) AS NPlayers FROM Player GROUP BY Country;
```

Country	NPlayers
India	10
Australia	5
West Indies	1
Sri Lanka	3
Pakistan	2
Zimbabwe	1
South Africa	4
New Zealand	2
England	1

Q18. Find the batting average of each player in the database. Present the PID of each Player.

mysql> SELECT PID,AVG(NRuns) AS AVGR FROM Batting GROUP BY PID;

```
MySQL localhost:3306 ssl anannya_match SQL > SELECT PID,AVG(NRuns) AS AVGR FROM Batting GROUP BY PID;
```

PID	AVGR
23001	19.0000
24001	42.0000
25001	53.5000
27001	4.5000
89001	91.0000
91001	60.0000
92002	1.0000
94002	17.0000
95001	1.0000
98002	2.5000
99001	7.0000
99002	2.0000

Solve the following

Q19. Find the batting average of each player from India in the database. Present first name of each player.

```
MySQL localhost:3306 ssl anannya_match SQL > SELECT FName , AVG(NRuns) as Batting_Average
```

```

-> from Player,Batting
-> where PlayerID = PID
-> and Country = "India"
-> group by PlayerID,FName
;

```

FName	Batting_Average
Yuvraj	19.0000
MS	53.5000
Praveen	4.5000
Sachin	91.0000
Harbhajan	2.5000

Q20. Find the average score for each player when playing against Australia .

```
MySQL localhost:3306 ssl anannya_match SQL > SELECT P.FName, B.PID, AVG(B.NRuns) AS Average_Score
-> FROM Player P
-> JOIN Batting B ON P.PlayerID = B.PID
-> JOIN ODIMatch M ON B.MatchID = M.MatchID
-> GROUP BY P.PlayerID, P.FName;OR M.Team2 = 'Australia')
-> GROUP BY P.PlayerID, P.FName;
```

FName	PID	Average_Score
Sachin	89001	91.0000
Yuvraj	23001	38.0000
MS	25001	36.0000
Adam	99002	2.0000
Ricky	95001	1.0000
Andrew	24001	42.0000
Breet	99001	7.0000
Praveen	27001	7.0000
Harbhajan	98002	3.0000

Q21. Find the ID of players that had a higher average score for all players when they played against Sri Lanka.

```
MySQL localhost:3306 ssl anannya_match SQL > SELECT B.PID, AVG(B.NRuns) AS Average_Score
-> FROM Batting B
-> JOIN ODIMatch M ON B.MatchID = M.MatchID
-> WHERE M.Team1 = 'Sri Lanka'
-> OR M.Team2 = 'Sri Lanka'
-> GROUP BY B.PID
-> ORDER BY Average_Score DESC;
```

PID	Average_Score
25001	71.0000
91001	60.0000
94002	17.0000
27001	2.0000
98002	2.0000
92002	1.0000
23001	0.0000